

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations

January 2023

18MEE52 – WELDING TECHNOLOGY

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Briefly explain the fusion welding and pressure welding.	4	L*1	1	1
	b) Explain: i) Straight and reverse polarity ii) Function of flux in SMAW.	6	L1	1	1
	c) Explain the HAZ microstructure in steel weldments with a help of iron carbon diagram.	10	L3	2	3
2.	a) Explain the importance of preheating, post heating and normalizing in welding.	10	L1	2	1
	b) Explain the principle of SMAW with a neat sketch. Briefly explain the need for constant current power source in SMAW.	10	L1	1	1
3.	a) Explain the principle of plasma arc welding and elucidate the significance of 'transfer mode' and 'non-transfer modes' in plasma arc welding.	10	L1	1	1
	b) What is hydrogen cracking in steel weldments? Explain briefly the conditions for hydrogen cracking.	10	L3	2	3
Unit – II					
4.	a) Explain liquid penetrant and magnetic particle test used to inspect weld joints with neat sketches.	10	L2	3	1
	b) With the help of neat sketch illustrate different types of distortion in welding.	10	L1	4	1
5.	a) What is x-ray radiography test? Explain with a neat sketch.	10	L1	3	1
	b) With the help of neat sketches explain 1G,2G,3G,4G,5G positions in groove welding	10	L1	4	1
6.	a) Explain the condition for formation of following defects in weldments (i) porosity (ii) hot cracks (iii) cold cracks (iv) undercut	10	L1	3	1
	b) What are the design considerations to be made to get good quality welds in groove, fillet and corner joints?	10	L1	4	1
Unit – III					
7.	a) How are fumes formed during welding? What are the factors affecting? Write briefly about general safety precautions to be taken during welding.	10	L1	5	1
	b) What is machine welding? List the advantages and limitations of automatic welding.	10	L1	5	1
8.	a) Write briefly about the application of welding in automobile industry.	10	L1	5	1
	b) Write short notes on i) electrical safety and precautions ii) 5s practice	10	L1	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
